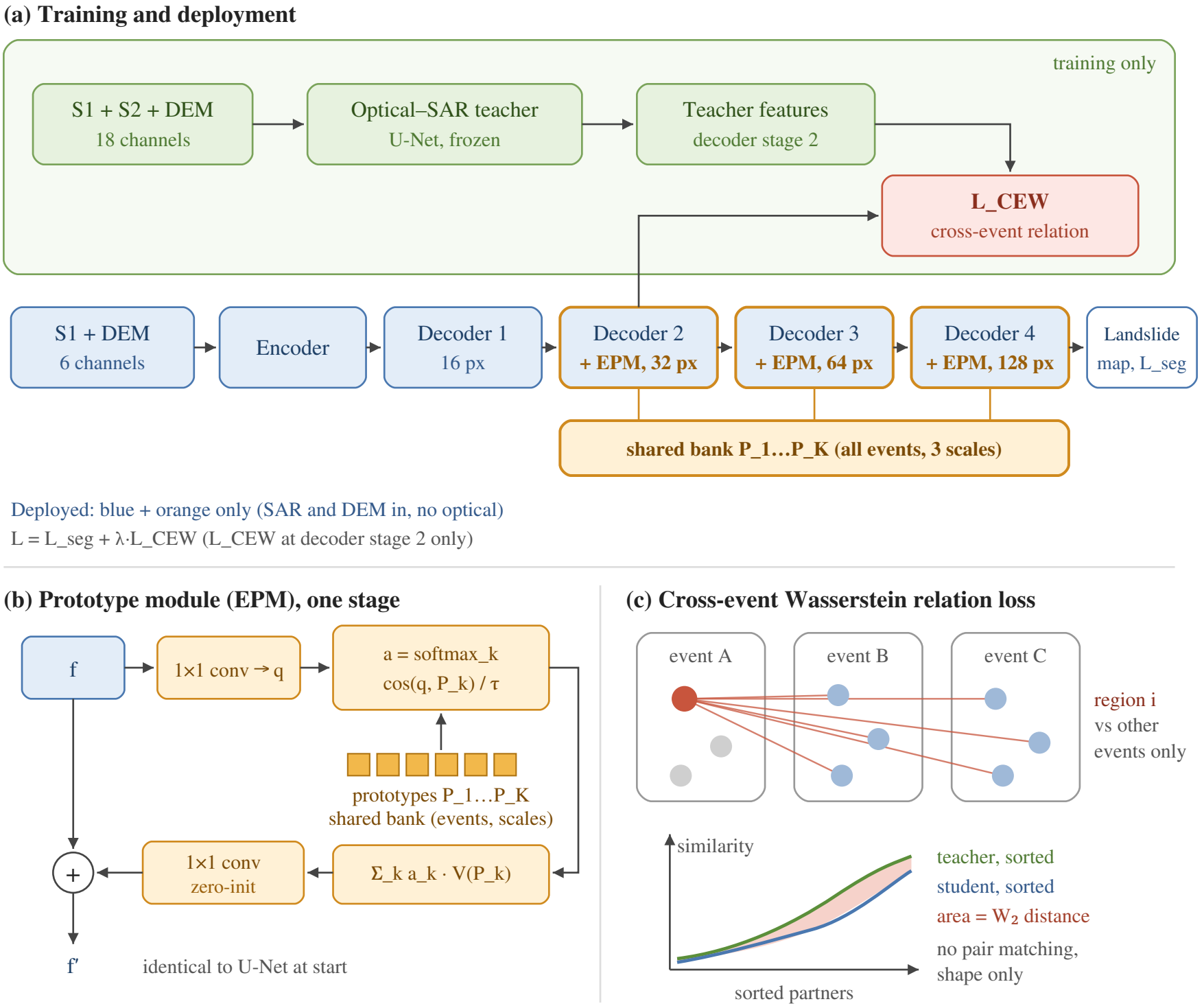

(a) Training and deployment
training only
S1 + S2 + DEM
18 channels
Optical–SAR teacher
U-Net, frozen
Teacher features
decoder stage 2
L_CEW
cross-event relation
S1 + DEM
6 channels
Encoder
Decoder 1
16 px
Decoder 2
+ EPM, 32 px
Decoder 3
+ EPM, 64 px
Decoder 4
+ EPM, 128 px
Landslide
map, L_seg
shared bank P_1…P_K (all events, 3 scales)
Deployed: blue + orange only (SAR and DEM in, no optical)
L = L_seg + λ·L_CEW (L_CEW at decoder stage 2 only)
(b) Prototype module (EPM), one stage
f
1×1 conv → q
a = softmax_k
cos(q, P_k) / τ
prototypes P_1…P_K
shared bank (events, scales)
+
1×1 conv
zero-init
Σ_k a_k · V(P_k)
f'
identical to U-Net at start
(c) Cross-event Wasserstein relation loss
event A
event B
event C
region i
vs other
events only
similarity
sorted partners
teacher, sorted
student, sorted
area = W₂ distance
no pair matching,
shape only